

QUANTUM COMPUTING • LEARN BY DOING

From Prediction to Experiment

Make a prediction first — then build the circuit and see whether reality agrees.

13 Make a Prediction

Quantum computing becomes much easier to understand when you stop thinking of a circuit as something you simply calculate and start treating it like an experiment.

Before running a circuit, ask yourself: “**What do I expect to happen when I measure it?**”

Key idea:

In quantum computing, prediction comes before measurement. We use the quantum state and the gates to predict the probabilities of possible measurement outcomes.

13.1 Start with the simplest circuit

Suppose we begin with one qubit in the state $|0\rangle$ and immediately measure it.



There are no gates changing the qubit, so the state remains $|0\rangle$.

$$P(0) = 1 \quad P(1) = 0$$

Make your prediction

If you run this circuit 100 times, how many times do you expect to observe each result?

0 → about 100 times

1 → about 0 times

13.2 Add an X gate

Now apply an **X gate**. We learned that X flips the computational-basis state.

$|0\rangle$

X

M

$$X|0\rangle = |1\rangle$$

Therefore, immediately before measurement, the qubit is $|1\rangle$.

$$P(0) = 0 \quad P(1) = 1$$

Predict before running

If this circuit is run 100 times, what should happen?

☐ Mostly 0

☐ Mostly 1

☐ 50% each

Answer: You should get 1 every time in an ideal, noiseless experiment.

13.3 Now make it interesting: the H gate

Start with $|0\rangle$ and apply the Hadamard gate.

 $|0\rangle$

H

M

The H gate changes $|0\rangle$ into an equal superposition:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

The amplitudes are:

Amplitude of $|0\rangle$

$$1 / \sqrt{2}$$

Amplitude of $|1\rangle$

$$1 / \sqrt{2}$$

To turn amplitudes into probabilities, square their magnitudes:

$$|1/\sqrt{2}|^2 = 1/2 = 50\%$$

$$P(0) = 50\% \quad P(1) = 50\%$$

13.4 Predict the experiment

Imagine running the circuit 100 times. Because each measurement has a 50% chance of producing either result, we expect the counts to be roughly balanced.

Possible result	Probability	Expected count out of 100
0	50%	About 50
1	50%	About 50

Important: “About 50” does not mean exactly 50. Quantum measurement is probabilistic, so one experiment might produce 47 zeros and 53 ones, while another might produce 52 zeros and 48 ones.

13.5 Prediction vs certainty

A quantum prediction usually tells us the **probabilities** of different outcomes rather than the exact result of one measurement.

Situation	Prediction
$ 0\rangle \rightarrow \text{Measure}$	0 with 100% probability
$ 1\rangle \rightarrow \text{Measure}$	1 with 100% probability
$ 0\rangle \rightarrow H \rightarrow \text{Measure}$	0 or 1, each with 50%

13.6 A useful habit

① Look

Look at the initial state and the gates in the circuit.

② Predict

Work out the final state and the measurement probabilities.

③ Test

Run the circuit and compare the observed counts with your prediction.

Think like a quantum experimenter:

Don't immediately run the circuit. First make a prediction. Then use the measurement results to test your understanding.

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Build It Yourself

Now it is your turn to create a quantum circuit from scratch.

The goal is not just to remember what the X and H gates do. The goal is to combine them, predict the result, and then test your prediction.

Challenge:

Build a circuit that starts with $|0\rangle$, uses quantum gates, and produces a predictable measurement result.

14.1 Your basic building blocks

At this stage, we can work with three essential components:

Component	Purpose
$ 0\rangle$	Initial state of the qubit
X	Flips $ 0\rangle \leftrightarrow 1\rangle$
H	Creates an equal superposition from $ 0\rangle$ or $ 1\rangle$
M	Measures the qubit

14.2 Build a circuit for $|1\rangle$

Your first task is simple:

Your task

Start with $|0\rangle$ and build a circuit that always measures 1.

Ask yourself: **Which gate changes $|0\rangle$ into $|1\rangle$?**

- 1
- Start with the qubit in $|0\rangle$.
- 2
- Apply an **X gate**.
- 3
- Measure the qubit.

$|0\rangle$

X

M

 $|0\rangle \rightarrow X \rightarrow |1\rangle \rightarrow M$

Prediction: The result should be 1 with probability 100%.

14.3 Build a 50–50 circuit

Now create a circuit that gives 0 and 1 with equal probability.

Your task

Start with $|0\rangle$ and use one gate to create a 50–50 measurement.

Which gate should you use?

Option A

X

Changes $|0\rangle$ directly into $|1\rangle$.

Option B

H

Creates an equal superposition.

$|0\rangle$

H

M

$$|0\rangle \rightarrow H \rightarrow (|0\rangle + |1\rangle)/\sqrt{2} \rightarrow M$$

Therefore:

$$P(0) = 50\% \quad P(1) = 50\%$$

14.4 Build something different

Here's a more interesting challenge.

Challenge

Start with $|0\rangle$, apply H, then apply H again. What do you think the final measurement will be?

☐ Always 0

☐ Always 1

☐ 50–50

Build the circuit:

 $|0\rangle$

H

H

M

We know:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Applying H once more gives:

$$H(H|0\rangle) = |0\rangle$$

So the complete transformation is:

$$|0\rangle \rightarrow H \rightarrow H \rightarrow |0\rangle$$

Prediction: The qubit should return to $|0\rangle$, so measurement gives **0 with 100% probability**.

14.5 Why did the second H undo the first?

The Hadamard gate is reversible. Applying it twice returns the qubit to its original state.

$$H^2 = I$$

Here, **I** is the identity operation — an operation that leaves the qubit unchanged.

Think of H like a reversible transformation

First H: $|0\rangle \rightarrow$ **superposition**

Second H: **superposition** $\rightarrow |0\rangle$

This is one of the first examples of how quantum gates can transform and later reconstruct a quantum state.

14.6 Build your own experiment

Now remove the instructions and design a circuit yourself.

Mini Design Challenge

Use only **X**, **H**, and **M**.

Try to create each of the following:

1. A circuit that always gives **0**.
2. A circuit that always gives **1**.
3. A circuit that gives **0 and 1 with equal probability**.
4. A circuit that uses at least **two gates**.

14.7 One possible set of solutions

Goal	Circuit	Expected result
Always 0	$ 0\rangle \rightarrow M$	0 = 100%

Always 1	$ 0\rangle \rightarrow X \rightarrow M$	1 = 100%
50–50	$ 0\rangle \rightarrow H \rightarrow M$	0 = 50%, 1 = 50%
Two gates, returns to 0	$ 0\rangle \rightarrow H \rightarrow H \rightarrow M$	0 = 100%

14.8 The experiment loop

This gives us a simple workflow that you can use for almost every circuit you learn from now on.

1. Build

Place the initial state and choose the gates.

2. Predict

Calculate the final state and expected probabilities.

3. Run

Measure many times and compare the actual counts with your prediction.

Build → Predict → Run → Compare → Understand

14.9 Optional: What this looks like in code

A quantum programming library can represent the same circuit using code. For example, conceptually:

```
qubit = |0>
H(qubit)
```

```
Measure(qubit)
```

The important thing is not the programming syntax yet. The important part is understanding what the circuit means:

$|0\rangle \rightarrow H \rightarrow \text{superposition} \rightarrow \text{measurement}$

14.10 What you should be able to do now

- ✓ Read a simple quantum circuit.
- ✓ Identify the starting state.
- ✓ Understand what X and H do.
- ✓ Predict measurement probabilities.
- ✓ Build a simple circuit yourself.
- ✓ Compare your prediction with experimental results.

The big idea:

Quantum computing is not just about performing calculations. It is about preparing a quantum state, transforming it with gates, predicting the probabilities, and then testing those predictions through measurement.

Quick recap:

Make a prediction → determine what outcomes you expect.

Build it yourself → create the circuit that should produce those outcomes.

Run it → measure many times and see how closely reality matches your prediction.